College of Medicine - HMU
Basic Science Department · 2nd Year Medical Students 2026

Determination of Serum Total Proteins



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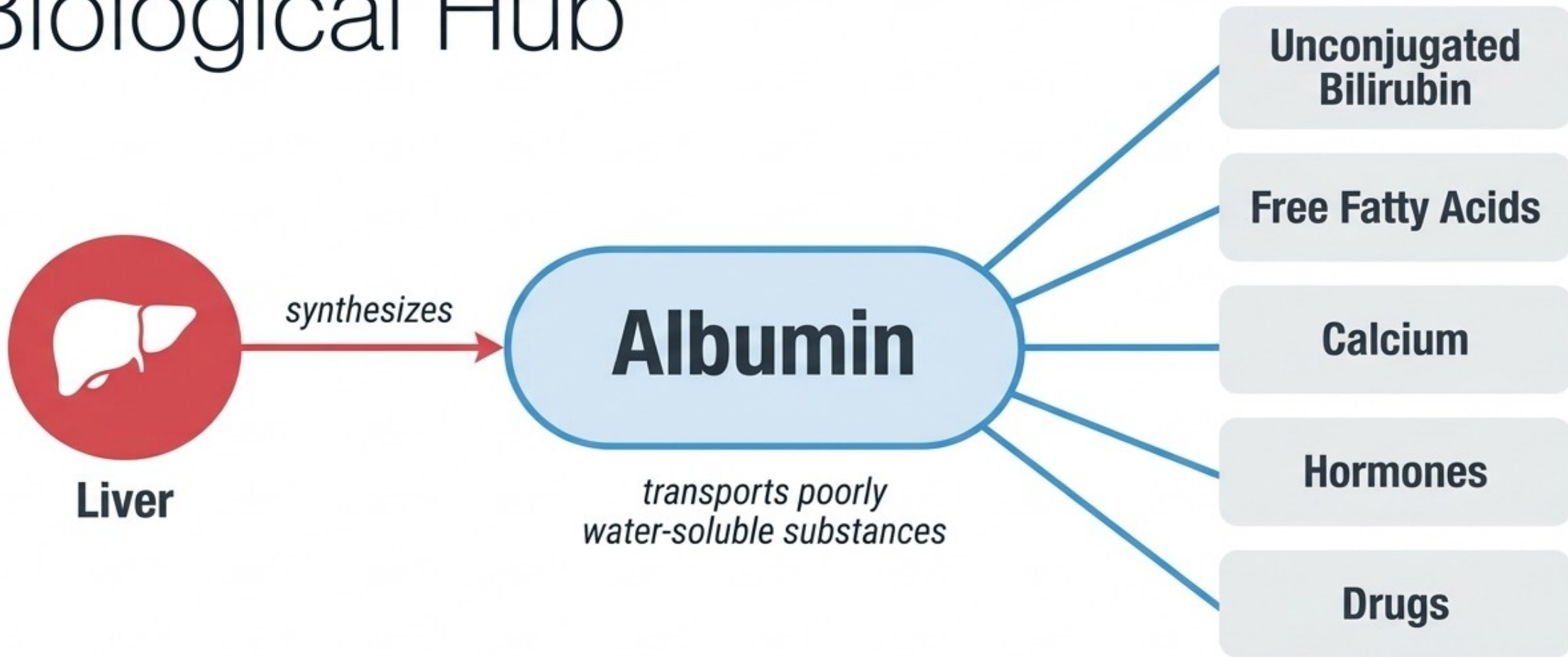
Learning Outcomes:

- Define serum total protein and distinguish serum from plasma proteins.
- Describe the major protein fractions and their principal physiological functions.
- State commonly used reference intervals for total protein and albumin, recognizing laboratory variation.
- Interpret major causes of hyperproteinemia and hypoproteinemia.
- the principles of the Biuret method for total protein and the bromocresol green (BCG) method for albumin.
- Calculate globulin concentration from total protein and albumin values.

Introduction

1. Total protein is a biochemical measurement of the combined concentration of proteins in serum or plasma.
2. Serum and plasma are not identical: serum is plasma after clotting and therefore lacks fibrinogen and most clotting factors.
3. The major circulating protein fractions include albumin and globulins; fibrinogen is an important plasma protein.
4. Globulins include several functional groups such as immunoglobulins, transport proteins, complement proteins, glycoproteins, and lipoproteins.
5. The liver synthesizes albumin and many other circulating plasma proteins. Immunoglobulins are produced by plasma cells.
6. Albumin is the major plasma protein and contributes substantially to colloid oncotic pressure.
7. Albumin also transports many poorly water-soluble substances, including unconjugated bilirubin, free fatty acids, calcium, several hormones, and many drugs.

Albumin as the Primary Biological Hub

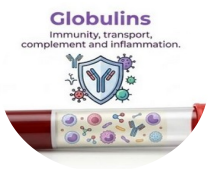


Gemini Notebook

Major Circulating Proteins



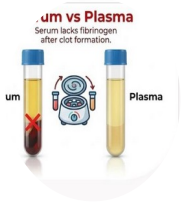
Albumin :Maintains oncotic pressure and transports many substances.



Globulins : Immunity, transport, complement and inflammation.



Fibrinogen: Important clotting protein present in plasma.



Serum vs Plasma: (Centrifuge / Separation) Serum lacks fibrinogen after clot formation.

The main functions of plasma proteins include:

1. Maintenance of colloid oncotic pressure and intravascular fluid balance.
2. Transport of hormones, lipids, bilirubin, metals, vitamins, and drugs.
3. Buffering of blood pH.
4. Contribution to nutritional amino-acid reserves during prolonged need.
5. Participation in coagulation, immunity, inflammation, and enzymatic regulation.
6. Blood coagulation: fibrinogen is converted to fibrin by thrombin during clot formation.
7. Protection: γ-globulins contain immunoglobulins (antibodies) that contribute to humoral immunity.
8. Regulatory and catalytic roles: plasma contains enzymes, protease inhibitors, complement proteins, and coagulation factors.

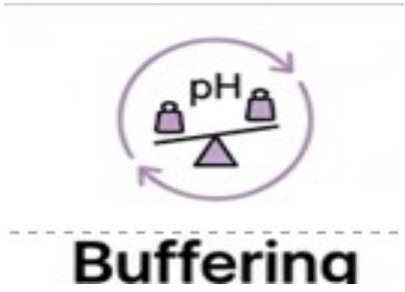
Functions of Plasma Proteins



Maintenance of colloid oncotic pressure and intravascular volume.



Moving hormones, lipids, bilirubin, metals, vitamins, and drugs.



Regulation and buffering of blood pH.



Fibrinogen conversion to fibrin by thrombin during clot formation.



γ-globulins (antibodies) contributing to humoral immunity.



Regulation – Plasma contains enzymes, protease inhibitors, and complement proteins.

Diagnostic Application:

Serum total protein and albumin help evaluate liver and kidney disease, nutritional status, hydration, and disorders involving immunoglobulins.

Abnormal results should be interpreted with the clinical context and, when indicated, serum protein electrophoresis.

Parameters	Reference Range
Total protein (serum)	6.3–7.9 g/dL
Albumin	3.5–5.0 g/dL
Globulins (calculated)	~2.0–3.5 g/dL
Fibrinogen (plasma)	~0.2–0.4 g/dL

Clinical Significance:

A. Hyperproteinemia:

Increased total protein is most often related to hemoconcentration or increased globulin production. Common settings include:

1. Monoclonal gammopathies, including multiple myeloma, a plasma-cell neoplasm that may produce large amounts of monoclonal immunoglobulin.

2. Chronic infections and chronic inflammatory or autoimmune disorders with polyclonal immunoglobulin elevation.

3. Dehydration, causing a relative increase in both albumin and total protein.

4. Marked hyperproteinemia should prompt consideration of serum protein electrophoresis and further investigation.

B-Hypoprotenemia

A decrease in total protein may result from low albumin, low globulins, or both. Major mechanisms include:

1. Decreased synthesis: advanced chronic liver disease and severe protein-energy malnutrition.
2. Reduced absorption or protein-losing gastrointestinal disease.
3. Excessive protein loss: nephrotic syndrome, extensive burns, protein-losing enteropathy, or major hemorrhage.
4. Hemodilution: overhydration can lower measured albumin and total protein concentrations.
5. Hypogammaglobulinemia can lower total protein even when albumin is not markedly reduced.

Principle:

1- Total Proteins:

Biuret method: in an alkaline medium, cupric (Cu^{2+}) ions react with peptide bonds to form a violet-colored complex. The color intensity is proportional to the protein concentration.

2- Albumin:

Bromocresol green (BCG) dye-binding method: at acidic pH (about 4.2), albumin binds the anionic dye to form a colored complex measured photometrically.

BCG may show positive bias in some clinical samples because of nonspecific binding.

Calculated globulin = Total protein − Albumin

References:

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5. MedlinePlus Medical Encyclopedia. Total protein; reviewed May 19, 2025.

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Thank You

Questions?